

Mutual Fund Analytics Platform

Data Analytics Internship Project Report

Submitted By

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Internship Organization

Bluestock Fintech

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Executive Summary

This project, "Mutual Fund Analytics Platform", was developed as part of the Data Analytics Internship at Bluestock Fintech. The primary objective was to analyze mutual fund performance, investor behavior, portfolio risk, and market trends using Python, SQL, and Power BI.

The project involved collecting, cleaning, and processing mutual fund datasets containing scheme performance, NAV history, investor transactions, SIP inflows, portfolio holdings, and benchmark indices. Extensive exploratory data analysis was performed to understand fund characteristics and investor patterns.

Several financial performance metrics including Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and Rolling Sharpe Ratio were calculated to evaluate risk-adjusted returns and downside risk. Investor cohort analysis and SIP continuity analysis were conducted to identify investor behavior patterns and potential investment risks.

An interactive Power BI dashboard consisting of four analytical pages was developed to visualize industry trends, fund performance, investor analytics, and SIP market insights. A simple recommendation engine was also built to suggest suitable mutual funds based on investor risk appetite.

The project demonstrates the practical application of data analytics, financial analysis, business intelligence, and data visualization techniques to generate meaningful insights for investors and fund management professionals.

Chapter 1: Project Overview

1.1 Introduction

The mutual fund industry has experienced significant growth in recent years, attracting investors from diverse financial backgrounds. With the increasing number of mutual fund schemes available in the market, investors often face challenges in evaluating fund performance, understanding risk levels, and selecting suitable investment options.

Data analytics plays a crucial role in transforming large volumes of financial data into meaningful insights. By leveraging analytical techniques, investors and financial institutions can make informed decisions based on historical performance, risk-adjusted returns, investor behavior, and market trends.

This project, titled "Mutual Fund Analytics Platform", was developed during the Data Analytics Internship at Bluestock Fintech. The project focuses on analyzing mutual fund datasets using Python, SQL, and Power BI to generate actionable insights regarding fund performance, risk metrics, investor activity, and portfolio concentration.

The project combines data engineering, exploratory data analysis, financial analytics, business intelligence, and visualization techniques to create a comprehensive mutual fund analytics solution.

1.2 Problem Statement

The mutual fund market contains a large number of schemes with varying levels of risk, returns, portfolio allocations, and investor participation. Investors often struggle to compare funds effectively and identify investment opportunities that align with their risk appetite.

Additionally, financial institutions require analytical tools to monitor investor behavior, evaluate portfolio risk, measure performance against benchmarks, and identify emerging market trends.

The primary problem addressed by this project is the lack of a unified analytical platform capable of integrating multiple mutual fund datasets and generating meaningful insights through data-driven analysis and visualization.

1.3 Project Objectives

The key objectives of this project are:

- To collect and process mutual fund datasets from multiple sources.
- To perform data cleaning and transformation for reliable analysis.
- To conduct exploratory data analysis on mutual fund performance and investor activity.
- To calculate financial performance metrics such as Alpha, Beta, Sharpe Ratio, Sortino Ratio, and Maximum Drawdown.
- To analyze fund risk using Value at Risk (VaR) and Conditional Value at Risk (CVaR).
- To perform investor cohort analysis and SIP continuity analysis.
- To develop a simple mutual fund recommendation model based on risk appetite.
- To evaluate portfolio concentration using the Herfindahl-Hirschman Index (HHI).
- To create interactive dashboards using Power BI for business intelligence reporting.
- To generate actionable insights for investors and fund management professionals.

1.4 Scope of the Project

The scope of this project includes the analysis of mutual fund performance data, NAV history, SIP inflows, investor transactions, portfolio holdings, and benchmark indices. The project covers data collection and preprocessing, financial performance evaluation, risk assessment and analytics, investor behavior analysis, portfolio concentration analysis, interactive dashboard development, and recommendation model development.

The project is limited to historical data analysis and does not provide real-time investment recommendations or live market predictions. However, the analytical framework developed can be extended in future versions to support real-time market data and advanced predictive analytics.

1.5 Technologies Used

Technology	Purpose
Python	Data analysis and financial calculations
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Matplotlib	Data visualization
SQLite	Database management
SQL	Data querying and transformation
Power BI	Dashboard development
Jupyter Notebook	Analysis and documentation
Git & GitHub	Version control and project management

1.6 Expected Outcomes

The expected outcomes of the project include:

- A cleaned and structured mutual fund database.
- Comprehensive performance and risk analysis reports.
- Investor behavior analytics and cohort insights.
- Interactive Power BI dashboards.
- Mutual fund recommendation functionality.
- Actionable business insights for investment decision-making.

The successful completion of this project demonstrates the application of data analytics techniques in the financial services domain and highlights the value of data-driven decision-making in mutual fund analysis.

Chapter 2: Dataset Description and Data Collection

2.1 Overview of the Dataset

The Mutual Fund Analytics Platform was developed using multiple datasets containing information related to mutual fund schemes, Net Asset Value (NAV) history, investor transactions, SIP inflows, portfolio holdings, benchmark indices, and industry-level statistics.

The datasets were collected, cleaned, transformed, and stored in a structured format to support exploratory analysis, performance evaluation, risk assessment, investor analytics, and dashboard development. The project utilized both raw datasets and processed datasets generated during the data preparation phase.

2.2 Dataset Sources

The project datasets were organized into two categories:

Raw Data

Raw datasets contain original information collected from various financial data sources before cleaning and transformation.

Processed Data

Processed datasets were generated after data cleaning, feature engineering, metric calculations, and aggregation operations. These processed datasets were used for analytical reporting, dashboard creation, and advanced financial analysis.

2.3 Description of Raw Datasets

1. Fund Master Dataset (01_fund_master.csv)

Contains master information for all mutual fund schemes. Key attributes: AMFI Code, Scheme Name, Fund House, Category, Plan Type, Risk Grade, Expense Ratio, Assets Under Management (AUM). Used as the primary reference table for identifying and categorizing mutual fund schemes.

2. NAV History Dataset (02_nav_history.csv)

Contains historical Net Asset Value (NAV) records. Key attributes: AMFI Code, Date, NAV. Used to calculate daily returns, risk metrics, rolling performance indicators, VaR, CVaR, and Sharpe Ratios.

3. AUM by Fund House Dataset (03_aum_by_fund_house.csv)

Provides Assets Under Management information by fund house. Key attributes: Fund House, AUM Value, Number of Schemes, Date. Used to analyze market share and industry-level fund distribution.

4. Monthly SIP Inflows Dataset (04_monthly_sip_inflows.csv)

Contains monthly Systematic Investment Plan (SIP) contribution statistics. Key attributes: Month, SIP Inflow Amount, New SIP Accounts, Active SIP Accounts. Used to study investor participation trends and SIP growth patterns.

5. Category Inflows Dataset (05_category_inflows.csv)

Records net inflows into different mutual fund categories. Key attributes: Category, Month, Net Inflow Amount. Used to analyze investor preferences across mutual fund categories.

6. Industry Folio Count Dataset (06_industry_folio_count.csv)

Contains the number of investor folios across different segments. Key attributes: Equity Folios, Debt Folios, Hybrid Folios, Total Folios. Used to evaluate industry growth and investor participation.

7. Scheme Performance Dataset (07_scheme_performance.csv)

Contains performance metrics and risk indicators for mutual fund schemes. Key attributes: Scheme Name, Returns (1Y/3Y/5Y), Alpha, Beta, Sharpe Ratio, Sortino Ratio, Standard Deviation, Maximum Drawdown, Morningstar Rating. Used for fund comparison and performance evaluation.

8. Investor Transactions Dataset (08_investor_transactions.csv)

Contains transaction-level investor information. Key attributes: Investor ID, Transaction Date, Transaction Type, Investment Amount, State, City, Age Group, Income Category. Used for investor cohort analysis and SIP continuity analysis.

9. Portfolio Holdings Dataset (09_portfolio_holdings.csv)

Contains portfolio allocation details of mutual funds. Key attributes: AMFI Code, Stock Symbol, Stock Name, Sector, Weight Percentage, Market Value. Used to evaluate portfolio diversification and calculate the Herfindahl-Hirschman Index (HHI).

10. Benchmark Indices Dataset (10_benchmark_indices.csv)

Contains benchmark index values for performance comparison. Key attributes: Index Name, Date, Closing Value. Used for benchmark analysis and fund performance comparison.

2.4 Data Collection Process

The data collection process involved gathering mutual fund information from structured CSV datasets and importing them into the project environment. The collected datasets were stored within a centralized project directory and subsequently loaded into Python using the Pandas library for preprocessing and analysis.

The data collection workflow consisted of:

1. Dataset acquisition and organization.
2. Data validation and integrity checks.
3. Missing value identification.
4. Duplicate record detection.
5. Data type correction and standardization.
6. Storage in structured formats for further analysis.

2.5 Data Preparation and Cleaning

Before performing analytical tasks, several preprocessing steps were applied: handling missing values, removing duplicate records, converting date fields into datetime format, standardizing categorical variables, creating derived metrics, calculating financial performance indicators, and preparing datasets for dashboard visualization.

2.6 Summary

The project utilized a comprehensive collection of mutual fund datasets covering scheme information, historical NAV values, investor activity, portfolio composition, and market benchmarks. These datasets provided the necessary foundation for performance analysis, risk assessment, investor analytics, and dashboard development.

Chapter 3: Data Preprocessing and Cleaning

3.1 Introduction

Data preprocessing is one of the most critical stages in the data analytics lifecycle. Raw datasets often contain missing values, duplicate records, inconsistent formats, and data quality issues that can affect the accuracy of analytical results. The objective of the preprocessing phase was to transform the raw mutual fund datasets into clean, structured, and analysis-ready datasets suitable for exploratory analysis, financial calculations, advanced analytics, and dashboard development. This phase was implemented primarily using Python, Pandas, NumPy, and SQLite.

3.2 Data Cleaning Workflow

The data preprocessing workflow followed these steps:

7. Data Loading
8. Data Inspection
9. Missing Value Handling
10. Duplicate Record Removal
11. Data Type Conversion
12. Feature Engineering
13. Data Validation
14. Export of Cleaned Datasets

3.3 Data Loading

All datasets were imported into Python using the Pandas library. After loading, the structure of each dataset was examined using `df.head()`, `df.info()`, and `df.describe()` to identify data quality issues and understand dataset characteristics.

3.4 Missing Value Handling

Several datasets contained missing or incomplete records. The following strategies were applied:

Numerical Columns

Missing numerical values were handled using mean imputation, median imputation, or zero replacement where appropriate.

Categorical Columns

Missing categorical values were replaced using the most frequent category or an 'Unknown' placeholder. This ensured that no important records were lost due to missing values.

3.5 Duplicate Record Removal

Duplicate records were identified and removed using `df.drop_duplicates()` across fund datasets, investor transactions, portfolio holdings, and NAV records. Removing duplicates improved dataset consistency and reliability.

3.6 Date Standardization

Several datasets contained date fields requiring conversion into proper datetime format using `pd.to_datetime()`. Date standardization enabled time-series analysis, rolling return calculations, cohort analysis, SIP continuity tracking, and dashboard trend visualization.

3.7 Data Type Conversion

Incorrect data types were identified and corrected by converting numerical fields using `pd.to_numeric()` and date fields using `pd.to_datetime()`. Proper data types improved processing speed and analytical accuracy.

3.8 Feature Engineering

Daily Return Calculation

Daily returns were derived from NAV history using the formula: $\text{Return}_t = (\text{NAV}_t - \text{NAV}_{(t-1)}) / \text{NAV}_{(t-1)}$. These were used for VaR, CVaR, Rolling Sharpe Ratio, and Risk Analysis.

Investor Cohort Year

A cohort year was created using the investor's first transaction date to support cohort analysis and investor segmentation.

SIP Continuity Features

Additional features created include: number of SIP transactions, average gap between SIPs, and at-risk investor flag. These were used for identifying investors with irregular investment behavior.

3.9 Data Validation

After cleaning, validation checks were performed including record count verification, null value checks, duplicate checks, and data type verification to ensure datasets were ready for analysis.

3.10 Processed Datasets Generated

Processed Dataset	Purpose
02_nav_history_clean.csv	NAV analysis
07_scheme_performance_clean.csv	Performance analytics
08_investor_transactions_clean.csv	Investor analytics
returns_computed.csv	Return calculations
sharpe_report.csv	Risk-adjusted performance
sortino_values.csv	Downside risk analysis
max_drawdown.csv	Drawdown analysis
var_cvar_report.csv	Risk analytics

3.11 Benefits of Data Cleaning

The preprocessing phase provided: improved data accuracy, reduced analytical errors, consistent formatting across datasets, reliable financial calculations, better dashboard performance, and enhanced visualization quality.

3.12 Summary

Data preprocessing transformed raw mutual fund datasets into structured, high-quality analytical assets. Missing values were handled, duplicates removed, dates standardized, and new features engineered to support advanced analytics and dashboard development. The cleaned datasets enabled accurate performance evaluation, investor analysis, risk assessment, and business intelligence reporting.

Chapter 4: Exploratory Data Analysis (EDA)

4.1 Introduction

Exploratory Data Analysis (EDA) is the process of examining datasets to understand their structure, identify patterns, detect anomalies, and generate insights before performing advanced analytics. In this project, EDA was conducted on mutual fund schemes, Assets Under Management (AUM), SIP inflows, category-wise investments, investor participation, portfolio holdings, and benchmark indices.

4.2 Objectives of Exploratory Data Analysis

- To understand the distribution of mutual fund assets.
- To analyze industry growth and investor participation.
- To identify leading fund houses and schemes.
- To evaluate SIP contribution trends.
- To analyze category-wise investment flows.
- To identify relationships between fund characteristics and performance.
- To prepare datasets for advanced financial analytics and dashboard reporting.

4.3 Assets Under Management (AUM) Analysis

AUM represents the total market value of investments managed by a mutual fund or fund house. The analysis focused on the largest mutual fund schemes by AUM, distribution of assets across schemes, and fund house market concentration.

Observations

- A small number of large schemes accounted for a significant portion of total industry assets.
- Equity-oriented funds generally maintained higher AUM values compared to niche categories.
- Several large-cap and diversified funds consistently ranked among the largest schemes.

AUM serves as an indicator of investor trust, market presence, and operational scale.

4.4 Monthly SIP Inflow Analysis

The monthly SIP inflow dataset was analyzed to evaluate growth in investor participation, monthly contribution trends, and changes in investment behavior over time.

Key Findings

- SIP inflows demonstrated consistent growth across the observed period.
- Retail participation remained strong despite market fluctuations.
- Higher SIP inflows indicated increasing investor confidence in long-term wealth creation.

4.5 Category-Wise Investment Flow Analysis

Mutual fund investments are distributed across Equity Funds, Debt Funds, Hybrid Funds, Liquid Funds, and Index Funds. The category inflow dataset was analyzed to understand investor preferences.

Observations

- Equity funds attracted the highest investment inflows.
- Debt and liquid funds remained popular among conservative investors.
- Hybrid funds provided balanced exposure and maintained steady investor interest.

4.6 Industry Folio Analysis

The folio dataset was used to analyze total investor participation, growth in mutual fund adoption, and segment-wise folio distribution.

Findings

- Equity funds accounted for the majority of investor folios.
- Total folio counts increased steadily, reflecting industry growth.
- Retail participation continued to drive market expansion.

4.7 Scheme Performance Exploration

An initial exploration of scheme returns was conducted examining 1-Year, 3-Year, and 5-Year returns.

Observations

- Mid-cap and small-cap funds generally delivered higher long-term returns.
- Large-cap funds offered more stable performance.
- Return variability increased with higher-risk categories.

4.8 Distribution of Daily Returns

Daily return distributions were generated using historical NAV data using the formula: $\text{Daily Return} = (\text{NAV}_t - \text{NAV}_{(t-1)}) / \text{NAV}_{(t-1)}$.

Statistical Summary

- Mean daily return close to zero.
- Moderate volatility levels.
- Presence of both positive and negative market movements.
- Limited extreme outliers.

The return distribution appeared realistic and suitable for financial risk analysis, essential for calculating Sharpe Ratio, Sortino Ratio, VaR, and CVaR.

4.9 Investor Demographic Analysis

Investor transaction data was analyzed using Age Group, Gender, Income Category, and Geographic Location.

Findings

- Investors were distributed across multiple income groups.
- Urban and metropolitan cities contributed a significant share of investments.
- Different age groups displayed varying investment patterns.

4.10 Geographic Distribution Analysis

Investment activity was evaluated across states, cities, and city tiers.

Observations

- Tier-1 cities contributed the highest transaction volumes.
- Tier-2 and Tier-3 cities demonstrated increasing participation.
- Geographic diversification indicated expanding market reach.

4.11 Correlation Analysis

Correlation analysis was performed to identify relationships among financial variables including Returns, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Expense Ratio, and AUM.

Observations

- Positive relationships existed between certain return-based metrics.
- Risk measures exhibited varying degrees of association.
- Larger AUM did not necessarily guarantee higher returns.

4.12 Key Insights from EDA

15. Equity-oriented schemes dominated industry assets and investor participation.
16. SIP inflows demonstrated sustained growth, reflecting increasing investor confidence.
17. Mid-cap and small-cap funds generally delivered stronger long-term returns.
18. Investor participation expanded across multiple geographic regions and income groups.
19. Return distributions appeared stable and suitable for advanced financial modeling.
20. Larger funds did not always outperform smaller funds on a risk-adjusted basis.
21. Growing folio counts highlighted the continued expansion of the mutual fund industry.

4.13 Summary

Exploratory Data Analysis provided a comprehensive understanding of mutual fund industry trends, investor behavior, asset distribution, and fund performance characteristics. The findings established a strong analytical foundation for the next phase focused on advanced performance evaluation and financial risk analytics.

Chapter 5: Performance Analytics

5.1 Introduction

Performance Analytics focuses on evaluating mutual fund schemes based on return generation, risk-adjusted performance, benchmark comparison, and downside risk. Various financial metrics were calculated to identify top-performing funds and assess their effectiveness under different market conditions. The analysis included CAGR, Sharpe Ratio, Sortino Ratio, Alpha and Beta, Maximum Drawdown, Fund Scorecard, and Benchmark Comparison.

5.2 Compound Annual Growth Rate (CAGR)

CAGR measures the annualized growth rate of an investment over a specified period. Formula: $CAGR = (NAV_{end} / NAV_{start})^{(1/n)} - 1$, where n is the number of years.

Top Performing Funds (3-Year CAGR)

Fund Name	CAGR (%)
Axis Midcap Fund - Regular - Growth	36.07
HDFC Mid-Cap Opportunities Fund - Regular - Growth	33.63
ABSL Frontline Equity Fund - Regular - Growth	32.53
Mirae Asset Large Cap Fund - Regular - Growth	31.28
ICICI Pru Midcap Fund - Regular - Growth	30.21

Mid-cap funds generated the highest long-term returns. Axis Midcap Fund achieved the best 3-year CAGR of 36.07%, indicating strong growth potential compared to other schemes.

5.3 Sharpe Ratio Analysis

Sharpe Ratio measures risk-adjusted returns by comparing excess return against total volatility. Formula: $\text{Sharpe} = (R_p - R_f) / \sigma_p$, where $R_f = 6.5\%$ (Risk-Free Rate). A higher Sharpe Ratio indicates better returns for each unit of risk taken.

Top Funds by Sharpe Ratio

Fund Name	Sharpe Ratio
Mirae Asset Large Cap Fund	1.45
Kotak Flexicap Fund	1.31
Mirae Asset Tax Saver Fund	1.23
SBI Bluechip Fund	1.21
ICICI Pru Midcap Fund	1.18

Mirae Asset Large Cap Fund delivered the highest risk-adjusted performance with a Sharpe Ratio of 1.45, demonstrating superior reward relative to volatility.

5.4 Sortino Ratio Analysis

The Sortino Ratio is a modified version of the Sharpe Ratio that considers only downside risk. Formula: $\text{Sortino} = (R_p - R_f) / \text{Downside Deviation}$. It focuses on harmful volatility rather than total volatility.

Top Funds by Sortino Ratio

Fund Name	Sortino Ratio
Mirae Asset Large Cap Fund	2.39
Kotak Flexicap Fund	2.36
Mirae Asset Tax Saver Fund	2.15

Fund Name	Sortino Ratio
SBI Bluechip Fund	2.14
ICICI Pru Midcap Fund	2.03

Mirae Asset Large Cap Fund provided the strongest downside-risk-adjusted performance among all analyzed schemes.

5.5 Alpha and Beta Analysis

Alpha measures a fund's excess return relative to its benchmark. Beta measures the sensitivity of a fund's returns relative to market movements. Linear regression was performed using fund returns and Nifty 100 benchmark returns.

Top Funds by Alpha

Fund Name	Alpha	Beta
SBI Small Cap Fund - Regular	0.303	-0.023
DSP Small Cap Fund	0.301	0.011
ICICI Pru Midcap Fund	0.293	0.001
Mirae Asset Tax Saver Fund	0.283	0.018
Kotak Flexicap Fund	0.273	-0.023

Positive Alpha values indicate that these funds outperformed their benchmark after accounting for market movements. ICICI Pru Midcap Fund demonstrated strong alpha generation while maintaining a beta close to zero.

5.6 Maximum Drawdown Analysis

Maximum Drawdown measures the largest peak-to-trough decline experienced by a fund. It quantifies downside risk during adverse market conditions.

Funds with Highest Drawdowns

Fund Name	Max Drawdown
SBI Small Cap Fund - Direct	-52.57%
Axis Small Cap Fund	-51.68%
ABSL Small Cap Fund	-35.45%
DSP Small Cap Fund	-31.17%
SBI Small Cap Fund - Regular	-28.71%

Small-cap funds experienced the highest drawdowns, highlighting their higher volatility and risk exposure.

5.7 Fund Scorecard

A composite score was developed to rank mutual funds using multiple performance indicators.

Weighting Methodology

Metric	Weight
3-Year Return Rank	30%
Sharpe Ratio Rank	25%
Alpha Rank	20%
Expense Ratio Rank	15%
Maximum Drawdown Rank	10%

Top Funds by Scorecard

Fund Name	Score
Mirae Asset Tax Saver Fund	100.00
ICICI Pru Bluechip Fund	98.07
Axis Bluechip Fund	92.21
SBI Bluechip Fund	85.03
UTI Nifty 50 Index Fund	83.87

5.8 Benchmark Comparison

Fund performance was compared against benchmark indices including Nifty 50 and Nifty 100 using Return Comparison, Alpha, Beta, and Tracking Error. Tracking Error = $\text{Std}(\text{Fund Return} - \text{Benchmark Return}) \times \sqrt{252}$. The computed Tracking Error was 0.2174, indicating that selected funds maintained performance patterns reasonably aligned with benchmark movements while still generating alpha.

5.9 Key Findings

- 22. Axis Midcap Fund generated the highest 3-year CAGR at 36.07%.
- 23. Mirae Asset Large Cap Fund delivered the strongest risk-adjusted performance based on Sharpe and Sortino Ratios.
- 24. Several small-cap funds generated strong alpha but experienced significant drawdowns.
- 25. The composite fund scorecard identified Mirae Asset Tax Saver Fund as the highest-ranked scheme.
- 26. Benchmark analysis indicated strong performance relative to market indices with manageable tracking error.

5.10 Summary

Performance Analytics provided a comprehensive evaluation of mutual fund returns, risk-adjusted performance, benchmark behavior, and downside risk. Metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and composite scoring helped identify top-performing funds and assess their suitability for different investor profiles.

Chapter 6: Advanced Analytics

6.1 Introduction

While traditional performance metrics provide valuable insights, advanced analytics techniques offer deeper understanding of risk exposure, investor behavior, portfolio concentration, and investment sustainability. This chapter covers Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation System, and Sector Concentration Analysis (HHI).

6.2 Historical Value at Risk (VaR)

Value at Risk (VaR) measures the maximum expected loss that a fund may experience over a specified time period at a given confidence level. For this project, Historical VaR was calculated using a 95% confidence level: $VaR(95\%) = 5\text{th Percentile of Daily Returns}$.

Sample Results

Fund	VaR (95%)
HDFC Top 100 Fund	-1.44%
HDFC Mid-Cap Opportunities Fund	-1.90%
ABSL Small Cap Fund	-2.60%

Funds with larger negative VaR values exhibited greater downside risk and higher exposure to adverse market movements.

6.3 Conditional Value at Risk (CVaR)

CVaR (also known as Expected Shortfall) measures the average loss experienced beyond the VaR threshold: $CVaR = \text{Mean}(\text{Returns} \leq VaR)$. CVaR provides a more comprehensive measure of tail risk by considering the severity of extreme losses.

Sample Results

Fund	CVaR (95%)
HDFC Top 100 Fund	-1.81%
HDFC Mid-Cap Opportunities Fund	-2.35%
ABSL Small Cap Fund	-3.25%

ABSL Small Cap Fund demonstrated the highest downside risk with an average loss exceeding 3% during the worst-performing market periods.

6.4 Rolling 90-Day Sharpe Ratio

Rolling Sharpe Ratio evaluates risk-adjusted performance over a moving 90-day window. Unlike static Sharpe Ratios, rolling analysis reveals how performance changes over time.

Findings

- Liquid funds maintained consistently high rolling Sharpe Ratios.
- Equity funds experienced higher fluctuations during volatile market periods.
- Some schemes demonstrated strong performance persistence over multiple periods.

6.5 Investor Cohort Analysis

Investor Cohort Analysis was performed to study investment behavior based on the year of an investor's first transaction. Investors were grouped by their first investment year and metrics including Average SIP Amount, Total Investment Amount, and Investor Participation were analyzed.

Results

Cohort Year	Average SIP Amount	Total Invested
2024	Rs. 107,423	Rs. 349.11 Crore
2025	Rs. 109,159	Rs. 3.05 Crore

The 2024 cohort contributed the largest investment volume and represented the most significant investor segment within the dataset.

6.6 SIP Continuity Analysis

The purpose of SIP Continuity Analysis was to identify investors showing signs of irregular contribution behavior. For investors with six or more transactions, the average gap between transactions was calculated. Investors with average gaps exceeding 35 days were classified as At Risk.

Results

Status	Count
At Risk	2,762
Healthy	188

Approximately 93.6% of eligible investors were classified as At Risk, indicating potential discontinuation risks. Financial institutions can use this analysis to implement targeted retention campaigns.

6.7 Mutual Fund Recommendation System

A simple recommendation engine was developed to assist investors in selecting suitable mutual funds based on risk appetite. Risk categories include Low Risk, Moderate Risk, High Risk, and Very High Risk. Funds were ranked according to Sharpe Ratio and Risk Grade.

Sample Recommendations - Low Risk

Fund	Sharpe Ratio
ICICI Pru Liquid Fund	7.68
Kotak Liquid Fund	6.18
ABSL Liquid Fund	5.14

6.8 Sector Concentration Analysis (HHI)

Portfolio concentration was measured using the Herfindahl-Hirschman Index (HHI). Formula: $HHI = \sum(\text{Weight}_i^2)$. Higher HHI values indicate greater concentration in a limited number of sectors or holdings.

Results

Fund	HHI
Axis Bluechip Fund	0.206
ABSL Small Cap Fund	0.201
SBI Small Cap Fund	0.175
UTI Nifty 50 Index Fund	0.175

Axis Bluechip Fund demonstrated the highest portfolio concentration. Highly concentrated portfolios may generate higher returns but carry greater sector-specific risks.

6.9 Key Insights from Advanced Analytics

- 27. Small-cap funds exhibited the highest downside risk based on VaR and CVaR metrics.
- 28. Liquid funds achieved the highest Sharpe Ratios due to lower volatility and stable returns.
- 29. The 2024 investor cohort contributed the majority of total investments.
- 30. SIP continuity analysis revealed that more than 93% of eligible investors were classified as At Risk.
- 31. Axis Bluechip Fund and ABSL Small Cap Fund demonstrated the highest portfolio concentration levels.
- 32. Rolling Sharpe analysis highlighted significant performance variation across market cycles.
- 33. The recommendation system successfully identified suitable funds based on investor risk preferences.

6.10 Summary

Advanced Analytics extended the project's capabilities beyond traditional performance measurement by incorporating risk modeling, investor behavior analysis, portfolio concentration assessment, and recommendation systems. The findings provided deeper insights into mutual fund risk exposure, investor engagement patterns, and portfolio structure.

Chapter 7: Power BI Dashboard and Visualization

7.1 Introduction

Data visualization is an essential component of modern analytics because it enables stakeholders to interpret complex datasets quickly and make informed decisions. A comprehensive Power BI dashboard was developed integrating multiple datasets and analytical outputs, providing visual insights into mutual fund performance, investor behavior, industry trends, and SIP analytics.

7.2 Dashboard Architecture

Data Sources Used

- Scheme Performance Dataset
- NAV History Dataset
- Monthly SIP Inflows Dataset
- Investor Transactions Dataset
- Portfolio Holdings Dataset
- Benchmark Indices Dataset

Tools Used

Tool	Purpose
Power BI Desktop	Dashboard Development
Power Query	Data Transformation
DAX	Calculated Measures
Python/Pandas	Data Preparation

7.3 Dashboard Design Principles

Simplicity

Visuals were kept clean and easy to understand.

Consistency

A consistent color palette, typography, and layout were maintained across all pages.

Interactivity

Users can filter and explore data dynamically through slicers and cross-filtering.

Business Relevance

Visualizations were selected based on their ability to communicate meaningful business insights.

7.4 Page 1: Mutual Fund Industry Overview

Objective: To provide a high-level overview of the mutual fund industry and key market indicators.

Visual Components

- KPI Cards: Total AUM, Total SIP Inflows, Number of Schemes, Total Fund Houses
- AUM by Fund House
- Category-wise Asset Distribution
- Monthly SIP Growth Trend
- Industry Folio Growth

Key Insights

- Industry AUM demonstrated significant scale and growth.
- Equity funds dominated overall market assets.
- SIP contributions continued to increase over time.
- Investor participation showed strong expansion across the industry.

7.5 Page 2: Fund Performance Analytics

Objective: To evaluate mutual fund performance using return and risk metrics.

Visual Components

- Fund Comparison Table (Scheme Name, Alpha, Beta, Sharpe Ratio, Sortino Ratio, Maximum Drawdown)
- Top Performing Funds by Sharpe Ratio
- Alpha Comparison
- Return Distribution
- Risk vs Return Scatter Plot

Key Insights

- Mirae Asset Large Cap Fund achieved the highest Sharpe Ratio.
- Several mid-cap funds generated strong long-term returns.
- Small-cap funds demonstrated higher volatility and drawdown risk.
- Positive alpha values indicated benchmark outperformance for selected schemes.

7.6 Page 3: Investor Analytics

Objective: To analyze investor demographics, investment behavior, and transaction activity.

Visual Components

- Demographic Analysis: Age Group Distribution, Gender Distribution, Income Category Analysis
- Geographic Analysis: State-wise Investment Distribution, City-wise Transaction Analysis, Tier-wise Participation
- Transaction Analytics: Total Transactions, Average Investment Amount, Investor Segmentation

Key Insights

- Investors originated from diverse demographic segments.
- Urban regions contributed the largest share of investments.
- Different income groups displayed varying investment patterns.
- Transaction activity reflected strong retail participation.

7.7 Page 4: SIP and Market Trends Dashboard

Objective: To analyze SIP contributions and market participation trends.

Visual Components

- KPI Cards: Total SIP Inflows, Active SIP Accounts, New SIP Registrations
- Monthly SIP Inflows Trend
- SIP Growth Rate
- Category-wise Net Inflows
- Investor Participation Trends

Key Insights

- SIP investments showed steady long-term growth.
- Investor participation remained resilient despite market fluctuations.
- Equity categories attracted the highest inflows.
- Growing SIP adoption highlighted increasing investor awareness.

7.8 Interactive Features

Slicers

Users can filter data by Fund House, Scheme Name, Risk Grade, Category, and Time Period.

Cross-Filtering

Selecting one visual automatically filters related visuals.

Drill-Down Analysis

Users can navigate from summary-level insights to detailed fund-level information.

Tooltips

Custom tooltips provide additional context when hovering over visuals.

7.9 Challenges Faced During Dashboard Development

- Understanding Power BI as a beginner.
- Managing relationships between multiple datasets.
- Creating effective visual layouts.
- Designing meaningful KPIs.

- Optimizing dashboard performance.

These challenges were addressed through iterative design improvements and data model optimization.

7.10 Summary

The Power BI dashboard served as the visualization layer of the Mutual Fund Analytics Platform, enabling users to interact with data, explore trends, and gain actionable insights. By integrating industry metrics, performance analytics, investor behavior, and SIP trends into a unified interface, the dashboard significantly enhanced the accessibility and practical value of the analytical results.

Chapter 8: Conclusion and Future Enhancements

8.1 Conclusion

The Mutual Fund Analytics Platform was successfully developed as part of the Data Analytics Internship at Bluestock Fintech. The project demonstrated the practical application of data analytics, financial analysis, business intelligence, and visualization techniques within the mutual fund industry.

The project involved the collection, cleaning, processing, and analysis of multiple mutual fund datasets. Through systematic data preparation and exploratory analysis, valuable insights were generated regarding industry trends, investor behavior, risk exposure, and fund performance.

Several performance metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to evaluate mutual fund effectiveness. Advanced analytics techniques such as VaR, CVaR, Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation System, and HHI further enhanced the analytical capabilities of the platform.

A four-page interactive Power BI dashboard was developed to visualize industry trends, fund performance, investor analytics, and SIP market insights. The project successfully achieved its objectives by integrating data engineering, financial analytics, and business intelligence into a comprehensive mutual fund analysis solution.

8.2 Project Achievements

Data Analytics

- Successfully processed and analyzed multiple mutual fund datasets.
- Performed comprehensive exploratory data analysis.
- Generated actionable insights from financial and investor data.

Financial Analytics

- Calculated risk-adjusted performance metrics.
- Evaluated fund performance against benchmark indices.
- Assessed downside risk using VaR and CVaR techniques.

Investor Analytics

- Performed investor cohort analysis.

- Identified at-risk investors through SIP continuity analysis.
- Analyzed demographic and geographic investment patterns.

Business Intelligence

- Designed and developed an interactive Power BI dashboard.
- Created KPI-driven visualizations and reports.
- Enabled dynamic exploration of mutual fund data.

Recommendation System

- Developed a simple fund recommendation engine based on risk appetite and Sharpe Ratio rankings.

8.3 Key Learnings

Technical Skills

- Python Programming, Pandas, and NumPy
- Data Cleaning and Transformation
- Financial Analytics and SQL Queries
- Power BI Dashboard Development and Data Visualization

Analytical Skills

- Risk Assessment and Performance Evaluation
- Investor Behavior Analysis
- Portfolio Diversification Analysis
- Business Intelligence Reporting

Professional Skills

- Problem Solving and Project Planning
- Data Interpretation and Documentation
- Presentation Development

8.4 Limitations of the Project

Historical Data Dependence

The analysis is based entirely on historical data and does not guarantee future performance.

Limited Market Factors

Macroeconomic variables such as inflation, interest rates, and economic indicators were not incorporated.

Simplified Recommendation Model

The recommendation system uses risk grade and Sharpe Ratio but does not consider individual investor goals, investment horizon, or portfolio allocation strategies.

Static Data Environment

The project uses static datasets and does not integrate real-time market feeds.

8.5 Future Enhancements

Real-Time Data Integration

- Integrate live mutual fund NAV data.
- Connect with financial APIs for real-time updates.
- Provide continuously updated dashboards.

Machine Learning Models

- Fund return prediction models.
- Risk forecasting and investor churn prediction.
- SIP discontinuation prediction.

Personalized Recommendation Engine

- Investor profile-based recommendations.
- Goal-based investment planning.
- Portfolio optimization algorithms.

Advanced Risk Analytics

- Monte Carlo Simulation.

- Stress Testing and Scenario Analysis.
- Factor-Based Risk Models.

Web Application Deployment

- Develop a web-based analytics portal.
- Enable user login and personalized dashboards.
- Provide interactive investment analysis tools.

8.6 Final Remarks

The Mutual Fund Analytics Platform demonstrates how data analytics can be effectively applied within the financial services industry to transform raw data into meaningful insights. By combining statistical analysis, financial modeling, business intelligence, and visualization techniques, the project provides a comprehensive framework for evaluating mutual fund performance and investor behavior.

The successful completion of this project highlights the importance of data-driven decision-making in modern finance and showcases the practical skills acquired during the Data Analytics Internship at Bluestock Fintech.

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